

RQ4 - Network Analysis

COMP3020 Assessment 2

Research Question

RQ4: How is the direct-reply network structured, and which participants occupy important positions within it?

Network construction and analytical approach

A directed author-level reply network was constructed from the cleaned Hacker News dataset. Each node represents a unique Hacker News author. A directed edge from author A to author B indicates that A directly replied to a comment written by B. The graph is directed because reply interactions have a clear direction.

The `parent_id` of each comment was matched to the corresponding `comment_id` to identify the author of the parent comment. Top-level comments whose parent was the Hacker News story rather than another comment were excluded because they do not represent a direct author-to-author reply. Repeated replies between the same author pair were represented by one structural edge, while reply frequency was retained as an edge weight.

The analysis used node and edge counts, network density, connected components, in-degree centrality, and betweenness centrality. In-degree identifies authors who attract replies from many distinct users. Betweenness identifies authors occupying bridging positions between different parts of the observed reply structure.

Network structure

Metric	Result
Comments in clean dataset	278
Direct comment-to-comment replies	198
Authors in reply network	144
Unique directed author pairs	193
Network density	0.0094
Weakly connected components	9
Largest weak component	128 authors
Largest weak component share	88.9%
Strongly connected components	102
Largest strong component	10 authors

The network contained **144 authors and 193 unique directed author-to-author edges**. Its density was approximately **0.0094**, indicating a sparse interaction structure. Fewer than 1% of all possible directed author-to-author relationships were observed.

Although the network contained **nine weakly connected components**, the largest component contained **128 of the 144 authors (88.9%)**. Thus, despite overall sparsity, most participants belonged to one broadly connected conversational structure.

Centrality results

Author	In-degree	Betweenness
simonw	14	0.0191
fishtoaster	14	0.0030
dakshgupta	7	0.0141
logicchains	7	0.0089
apical_dendrite	7	0.0027
Swizec	6	0.0033
jryio	5	0.0005
duskdozer	4	0.0000

The highest in-degree values were recorded for **simonw and fishtoaster (14 each)**, indicating that they attracted replies from the greatest number of distinct participants. **simonw** also had the highest betweenness centrality (approximately **0.0191**), followed by **dakshgupta** (approximately **0.0141**) and **logicchains** (approximately **0.0089**). This suggests that **simonw** combined prominence as a reply target with a comparatively important bridging position.

In-degree distribution

Most participants have relatively low in-degree values, whereas a small number receive replies from considerably more distinct users. This indicates that interaction is unevenly distributed rather than shared equally across participants.

Full network visualisation

The full network is visually consistent with the density result: it is sparse overall, but contains a dominant connected structure. Larger nodes indicate participants who attract replies from more distinct authors.

Justified subgraph: largest weakly connected component

The largest weakly connected component was selected because it represents the main connected conversational structure rather than an arbitrary subset. It contains **128 authors, or 88.9% of all network participants**, while excluding smaller disconnected exchanges.

Answer to RQ4

RQ4 asked how the direct-reply network was structured and which participants occupied important positions within it. The results show that the network was **sparse, with a density of approximately 0.0094, but largely connected at the weak-component level**. Although nine weakly

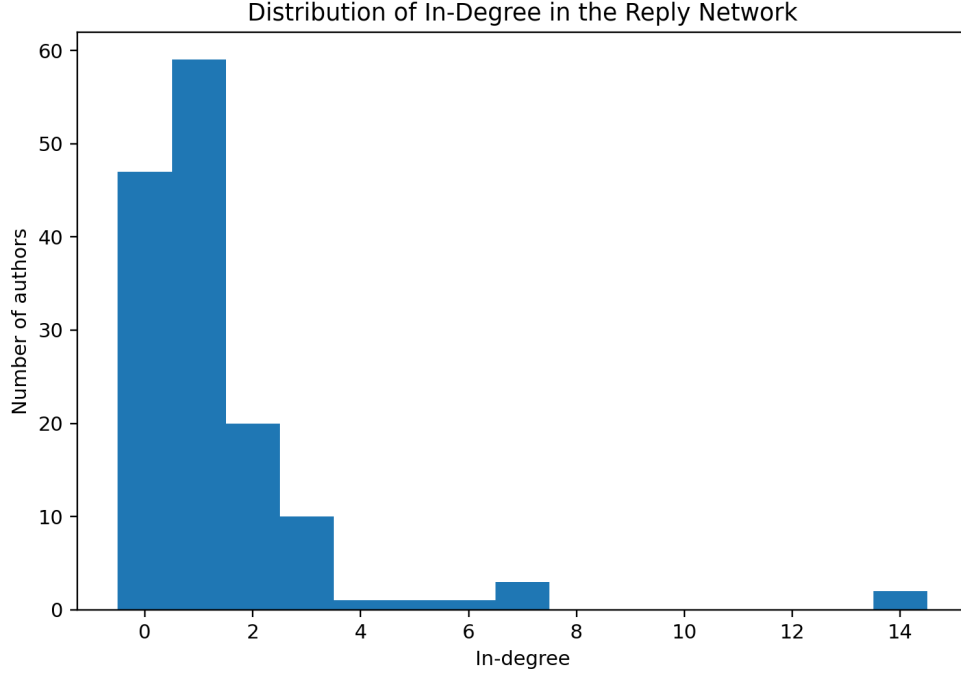


Figure 1: Distribution of in-degree in the direct-reply network.

connected components were identified, the largest contained **128 of 144 authors (88.9%)**, showing that most participants belonged to one broad conversational structure.

Interaction was also unevenly distributed. **simonw and fishtoaster had the highest in-degree centrality (14 each)**, indicating that they attracted replies from the largest number of distinct users. **simonw also had the highest betweenness centrality**, suggesting a bridging position between different areas of the observed discussion network. Overall, the selected discussions were characterised by a sparse but largely connected reply structure in which a relatively small number of participants occupied prominent reply-target or bridging positions.

Limitations

The network is based only on the selected Hacker News discussions and therefore does not represent the complete Hacker News community or all discussions of AI-generated code. Top-level comments do not generate author-to-author edges because their parent is the story rather than another participant’s comment, so the network captures direct commenter-to-commenter replies rather than all participation. The selected threads also differ in size and reply activity, meaning larger discussions may contribute disproportionately to the combined network. Finally, centrality measures represent structural prominence only and should not be interpreted as evidence of expertise, agreement, persuasion, or real-world influence.

Hacker News Direct-Reply Network

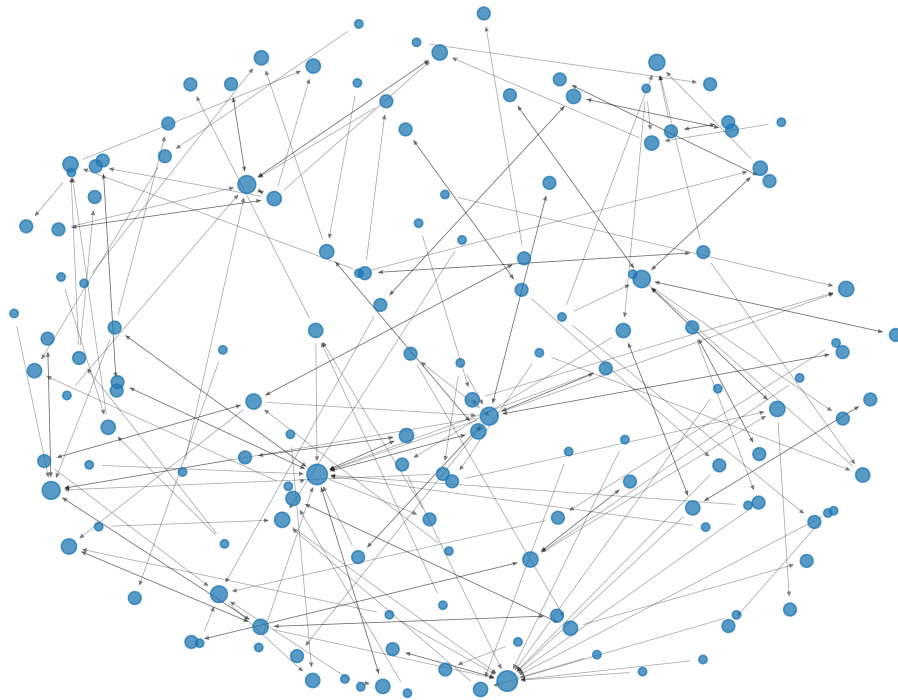


Figure 2: Full Hacker News direct-reply network. Node size represents in-degree.

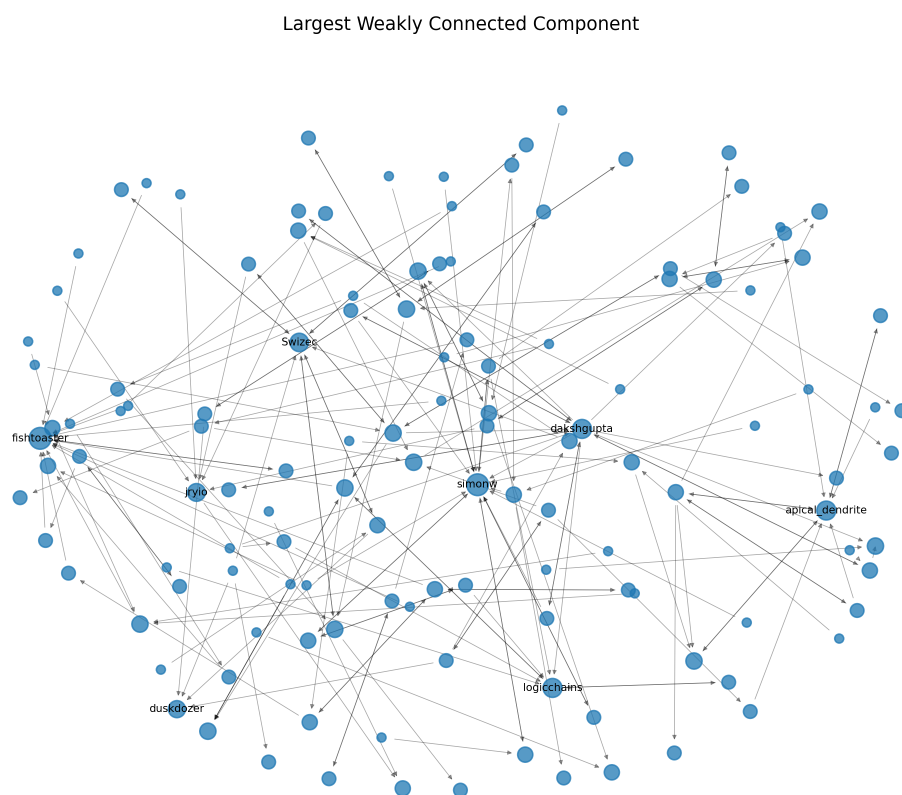


Figure 3: Largest weakly connected component. Labels identify relatively prominent reply targets.